

Theory of thermal unimolecular reactions at high pressures

Cite as: J. Chem. Phys. **75**, 226 (1981); <https://doi.org/10.1063/1.441829>

Published Online: 31 August 1998

J. Troe



View Online



Export Citation

ARTICLES YOU MAY BE INTERESTED IN

[Theory of thermal unimolecular reactions at low pressures. II. Strong collision rate constants. Applications](#)

The Journal of Chemical Physics **66**, 4758 (1977); <https://doi.org/10.1063/1.433838>

[Theory of thermal unimolecular reactions at high pressures. II. Analysis of experimental results](#)

The Journal of Chemical Physics **83**, 1010 (1985); <https://doi.org/10.1063/1.449464>

[Theory of thermal unimolecular reactions at low pressures. I. Solutions of the master equation](#)

The Journal of Chemical Physics **66**, 4745 (1977); <https://doi.org/10.1063/1.433837>



Time to get excited.

Lock-in Amplifiers – from DC to 8.5 GHz



Find out more



Zurich
Instruments

Theory of thermal unimolecular reactions at high pressures

J. Troe

Institut für Physikalische Chemie, Universität Göttingen, Tammannstrasse 6, D-3400 Göttingen, West Germany

(Received 5 December 1980; accepted 24 February 1981)

A simplified version of a statistical adiabatic channel model for unimolecular simple bond fission and the reverse radical association reactions is presented. The pattern of threshold energies for individual channels is analyzed and expressed empirically. High pressure rate constants for thermal dissociation and recombination are given in explicit form and applied to practical examples. The present theory interpolates between rigid and loose activated complex theory in a simple way with an account for angular momentum couplings. The interpolation parameter is the ratio α/β of the Morse parameter β and the looseness parameter α .

I. INTRODUCTION

In a previous series of articles¹⁻³ we have presented a reformulation of the theory of thermal unimolecular reactions at the low pressure limit. We have expressed the rate constant in a realistic and rational form to encourage the application to practical examples. In the present article we try to provide similarly simple expressions for the high pressure limit of thermal unimolecular reactions. We base our treatment on the statistical adiabatic channel model of unimolecular processes by Quack and Troe,⁴⁻⁶ of which a simplified version will be developed in the following. Our work applies in particular to *simple bond fission reactions* and the reverse *radical association reactions* (potential energy surface of the type II in the Waage-Rabinovitch nomenclature⁷). It does not provide much new information for complex unimolecular reactions with pronounced potential barriers in the forward and backward direction (potential surface of type I), for which traditional transition state theory like RRKM theory apparently works well.

The high pressure rate constant of a thermal unimolecular reaction k_∞ in statistical theories is expressed by

$$k_\infty = \frac{kT}{h} \frac{Q^*}{Q} \exp\left(-\frac{E_0}{kT}\right), \quad (1.1)$$

where E_0 is the threshold energy of the lowest reaction channel, Q is the partition function of the reactant molecule, and Q^* the partition function of the "activated complex." (Q and E_0 are expressed relative to the same energy zero level.) In the language of the statistical adiabatic channel model, Q^* is given⁴ by a pseudopartition function of the threshold energies E_{0i} of all individual channel states i :

$$Q^* = \sum_i \exp\left(-\frac{E_{0i} - E_0}{kT}\right). \quad (1.2)$$

Equation (1.2) can also be expressed by a total energy E , and total angular momentum J , dependent number of open channel states $W^*(E, J)$:

$$Q^* = \sum_{J=0}^{\infty} (2J+1) \int_0^{\infty} W^*(E, J) \exp\left(-\frac{E - E_0}{kT}\right) \frac{dE}{kT}. \quad (1.3)$$

Equations (1.2) and (1.3) illustrate the central problem of statistical theories, i.e., the *determination of the*

energy pattern of the channel threshold energies E_{0i} .

For complex unimolecular reactions (potential type I) all channel potential curves have their maximum near that configuration which corresponds to the maximum of the lowest channel. Then a common activated complex can be defined; Q^* corresponds to a partition function of activated complex "states." In this case, *rigid activated complex theory*, such as formulated by RRKM theory,⁸⁻¹⁰ calculates $W^*(E, J)$ from the state pattern of an assembly of harmonic oscillators (and free rotors). This approach provides satisfactory results for k_∞ as well as for the specific rate constants $k(E, J)$, although the assignment of the activated complex frequency pattern still has to be based on empirical guessing rules which, in the absence of accurate *ab initio* calculations, have to be calibrated against the experiments. Obviously, no unique frequency choice can be obtained in this way.¹¹ The application of rigid activated complex theory to simple bond fission reactions (potential type II) meets serious problems. The transformation of some oscillators of the reactant molecule into product rotors is accompanied by a considerable change in the state pattern. In addition, this transformation of the oscillators is governed by conservation of angular momentum with complicated coupling schemes between orbital motion and molecular rotations. Traditionally, the change of the state pattern is accounted for by loosening of oscillators, possibly including free rotors,⁸⁻¹⁰ in the activated complex structure. Corrections for the conservation of angular momentum, however, are difficult to apply.¹²

If, on the other hand, one assumes that the individual channel states open up at the energies of the states of the product molecules, conservation of total angular momentum can be accounted for exactly. This approach, the *phase space theory* or *loose activated complex theory*,¹³⁻¹⁶ again is characterized by an energy pattern composed of contributions from harmonic oscillators and free rotors. Unfortunately, this simple model generally provides an overestimate of the reaction rate when compared to experiments.¹⁷

In recent years different ways have been tried to bridge the gap between rigid and loose activated complex theory. Starting from the loose side, a hindrance parameter η has been defined¹⁸ which introduces steric hindrance of the free fragment rotations into a loose activated com-

plex theory. This hindrance parameter, which depends on the temperature, is a free parameter which is interpreted geometrically. In a different approach, in order to avoid the traditional, more or less arbitrary, guess of loosening activated complex frequencies, one may also try to represent the transformation of the reactant oscillators into product rotors in the spirit of the bond energy-bond order method.¹⁹ In this way the change, for example, of oscillator frequencies along the reaction coordinate is described, in a first approximation, by one characteristic parameter α which is related to Pauling's rule, Badger's rule, or the BEBO expressions. This approach initially²⁰ has been combined with minimum density of states criteria²¹ to find optimum positions of an activated complex along the reaction coordinate. As a result, it was shown that the positions of activated complexes in a microcanonical description depend on the energy; in a canonical description they depend on the temperature. In later work,^{4,22,23} it turned out that *minimum number of states criteria* are more adequate. However, the calculation of adiabatic channel potential curves and the determination of the individual channel threshold energies E_{0i} , such as proposed by the *statistical adiabatic channel model*,⁴ is numerically not more difficult, accounts exactly for angular momentum coupling, and avoids a minimization procedure.

In practical applications, often much less detailed information is required than provided by the detailed statistical adiabatic channel model. Therefore, simplified versions are desired. In Ref. 17, k_{∞} was obtained from a *maximum free energy criterion* which, after calculation of the free energy along the reaction coordinate, again leads to temperature dependent positions of activated complexes. In contrast to the interpolation of channel states between reactant and product eigenvalues, here the total free energy was interpolated. Although the correspondence is not exact, a "translation" between the interpolation parameters ($\gamma \approx 0.75\alpha$) was proposed. In a corresponding approach to $k(E, J)$ one would interpolate $W^*(E, J)$ between the rigid and loose limits and apply the minimum number of states criterion.

In the present work, we try to go one step further. We stay in the framework of the statistical adiabatic channel model, i.e., we avoid minimization procedures, and we analyze the energy pattern of the channel threshold energies E_{0i} in an empirical way. After separation of the coordinates, we enter this pattern into simple statistical theories like the RRKM theory. However, the coupling of angular momenta is neglected in this way, such that corrections have to be applied at the end. For loosening oscillators, the energy pattern of the channel threshold energies turns out to correspond neither to harmonic oscillators nor to free or hindered rotors. Therefore, unusual pseudopartition functions of the activated complex arise. The interpolation between rigid and loose activated complex theory is done with the single parameter α of the statistical adiabatic channel model. Since a standard value of α describes many experimental results with reasonable accuracy, the present model allows for a prediction of k_{∞} and of $k(E, J)$ in a rational way. The present article describes the cal-

culation of k_{∞} ; $k(E, J)$ will be discussed in a subsequent paper.²⁴

In the present work the properties of the potential surface are characterized by the Morse parameter β and the looseness parameter α . Furthermore, channel eigenvalues are interpolated by the use of simple exponential switching functions $\exp[-\alpha(q - q_e)]$. Clearly, this model presents a very rough description of the potential which in the future has to be checked or modified by comparison with realistic *ab initio calculations* of the potential surface. The statistical adiabatic channel model is by no means limited to the use of the present simple potential model. With improved potentials, an analysis of the energy pattern of the channel threshold energies E_{0i} should be made such as described in the present paper. This analysis would then allow one to express the connection between potential surface and rate constant in a most economic and transparent way.

II. STATISTICAL ADIABATIC CHANNEL MODEL

In the statistical adiabatic channel model,⁴ adiabatic channel potential curves are constructed for all individual rovibrational channel states. The channel potential $V_a(q)$ is given by the sum of the Morse potential $V(q)$ and the channel eigenvalue $E_a(q)$ as a function of the reaction coordinate q , which for simple bond fission reaction is a suitable interfragment distance

$$V_a(q) = V(q) + E_a(q) . \quad (2.1)$$

The Morse potential is given by

$$V(q) = D \{1 - \exp[-\beta(q - q_e)]\}^2 , \quad (2.2)$$

with

$$D = \Delta H_0^\circ + E_{zn} - E_{zp} , \quad (2.3)$$

where ΔH_0° is the reaction enthalpy at 0 K, E_{zn} is the zero point energy of the reactant molecule, and E_{zp} is the zero point energy of the fragments. The Morse parameter β is expressed by

$$\beta = \sqrt{\frac{f_{RC}}{2D}} \cong \sqrt{\frac{2\pi^2\mu}{Dh^2}} \epsilon_{RC} , \quad (2.4)$$

with the force constant f_{RC} of the bond which corresponds to the reaction coordinate. μ is the reduced mass of the fragments. Wherever a force field calculation has been made, f_{RC} is used directly from this analysis. Otherwise, in Eq. (2.4) the quantum ϵ_{RC} of the oscillator resembling most the separation of the fragments is chosen. The channel eigenvalue $E_a(q)$ is interpolated between the eigenvalues $E_a(q_e, n)$ of the reactant states $|n\rangle$ and the eigenvalues $E_a(\infty, p)$ of the product states $|p\rangle$ by means of a single interpolation parameter α :

$$E_a(q) = E_a(q_e, n) \exp[-\alpha(q - q_e)] + E_a(\infty, p) \times \{1 - \exp[-\alpha(q - q_e)]\} + E_{cent}(q) . \quad (2.5)$$

The centrifugal energy $E_{cent}(q)$ is calculated separately as

$$E_{cent}(q) = B_{cent}(q)P(P+1) , \quad (2.6)$$

with

$$P = J \exp[-\alpha(q - q_e)] + l[1 - \exp[-\alpha(q - q_e)]] \quad (2.7)$$

interpolating between the quantum numbers J for reactant total angular momentum and l for fragment orbital motion; the effective rotational constant $B_{\text{cent}}(q)$ of the separating fragments is given by the mean of the two small rotational constants $B(q)$ and $C(q)$ for the separating fragment pair at a fixed bending angle.

The model is completed by a correlation scheme specifying which reactant state $|n\rangle$ adiabatically is transformed into the product state $|p\rangle$. In this scheme, all known constants of the motion such as conservation of total angular momentum and conservation of the vibrational quantum numbers in the conserved oscillators of the separating fragments have to be built in. This correlation scheme has the form

$$\begin{array}{ccccc} |n\rangle & \longleftrightarrow & |a\rangle & \longleftrightarrow & |p\rangle \\ J & \longleftrightarrow & J & \longleftrightarrow & J, \\ M & \longleftrightarrow & M & \longleftrightarrow & M, \\ v_{1n} & \longleftrightarrow & v_1 & \longleftrightarrow & v_{1p}, \\ v_{2n} & \longleftrightarrow & v_2 & \longleftrightarrow & v_{2p}, \\ \vdots & & \vdots & & \vdots \\ v_{rn} & \longleftrightarrow & v_r & \longleftrightarrow & v_{rp}, \end{array}$$

$$(v_{1n}^*, \dots, v_{rn}^*, \tau_n) \longleftrightarrow w \longleftrightarrow (j_1, m_1, \tau_1, j_2, m_2, \tau_2, l, m_l)$$

for nonlinear asymmetric top molecules (rotational quantum numbers J, M, τ_n) dissociating into two asymmetric top fragments (individual rotational quantum numbers j_1, m_1, τ_1 and j_2, m_2, τ_2), with quantum numbers l and m_l for orbital motion. Without too much difficulty, it is often possible to identify "conserved" reactant oscillators (quantum numbers $v_{1n}, \dots, v_{rn}$; possibly including internal rotors) which correspond to the product oscillators (quantum numbers $v_{1p}, \dots, v_{rp}$; possibly including internal rotors) and whose quantum numbers during the final adiabatic separation of the fragments are conserved individually. The remaining "disappearing" oscillators of the reactant molecule (quantum numbers $v_{1n}^*, \dots, v_{rn}^*$; one oscillator corresponding to the reaction coordinate q is missing) couple with the rotation and are transformed into product rotors such that only one common channel state quantum number w can be defined. w numbers these states in the order of increasing energy and degeneracy avoiding crossing of the channel potential curves $V_a(q)$. In the construction of the correlation scheme, the coupling restrictions $|\tau_n| \leq J$, $|\tau_1| \leq j_1$, $|\tau_2| \leq j_2$, $|M| \leq J$, $|m_1| \leq j_1$, $|m_2| \leq j_2$, $j_1 + j_2 + 1 = J$, and $m_1 + m_2 + m_l = M$ have to be obeyed.

Table I gives the number b of disappearing oscillators or internal rotors (quantum numbers $v_{1n}^*, \dots, v_{rn}^*$) which are transformed into fragment rotations and orbital motion. For a nonlinear reactant molecule, we include in the number b for simplicity the external rotor with the quantum number K (for a symmetric top, and τ_n for an asymmetric top).

TABLE I. Number b of disappearing oscillators of the reactant (including one external rotor for a nonlinear molecule) which are transformed into product rotations and orbital motion.

Fragment 1	Fragment 2	b
Atom	Linear	2
	Nonlinear	3
Linear	Linear	4
	Nonlinear	5
Nonlinear	Nonlinear	6

III. THRESHOLD ENERGY PATTERN FOR DISAPPEARING OSCILLATORS

In the following simplified version of the statistical adiabatic channel model, the channel coordinates are separated in the expression for the total channel threshold energy E_{0i} :

$$E_{0i} = \sum_{j=1}^r \epsilon_{ji} + \sum_{m=1}^b \epsilon_{mi}^* + E_0(J). \quad (3.1)$$

$E_0(J)$ is the centrifugal barrier, i.e., the lowest threshold energy for a given J ; ϵ_{ji} are contributions from the conserved oscillators, and ϵ_{mi}^* are contributions from the b disappearing oscillators (including, for a nonlinear reactant, the external rotation with quantum number τ_n). The separation of the coordinates neglects coupling at first; corrections for this error are applied at the end. The energy pattern of the channel maxima in the following is calculated from Eqs. (2.1)–(2.7) for $J=0$, and for excitation of individual channel oscillators. Afterwards, E_{0i} is composed from the individual contributions ϵ_{ji} and ϵ_{mi}^* of the separated channel oscillators, and from the centrifugal barriers $E_0(J)$.

We first consider the threshold energy $E_0 \equiv E_0(J=0)$ of the lowest reaction channel [see Eq. (1.1)] for which no channel oscillators and no rotations are excited. For this case, the statistical adiabatic channel model predicts the appearance of "adiabatic zero point barriers" as given by the maxima of the lowest channel potential

$$V_a(q, J=0) = D\{1 - \exp[-\beta(q - q_e)]\}^2 + \Delta E_x \exp[-\alpha(q - q_e)] + E_{zp}, \quad (3.2)$$

where

$$\Delta E_x = E_{zn} - E_{zp} - \epsilon_{RC}/2. \quad (3.3)$$

For given values of the parameters $\Delta E_x/D$ and α/β , the maxima of Eq. (3.2) are easily evaluated numerically. For $\alpha/\beta \rightarrow 0$, the barriers $E_0 - D - E_{zp}$ approach ΔE_x ; for $\alpha/\beta \gtrsim 0.8$, they disappear. In Table II the adiabatic zero point barriers are expressed by $(E_0 - D - E_{zp})/\Delta E_x$. Exact calculations (first rows) are compared with an empirical representation (second rows)

$$\Delta E_{0x} \equiv E_0 - D - E_{zp} \approx \Delta E_x \exp[-C_1 \alpha/(\beta - 0.8\alpha)], \quad (3.4)$$

with

TABLE II. Adiabatic zero point barriers. The table gives $(E_0 - D - E_{zp})/\Delta E_s$ for various α/β and $\Delta E_s/D$. First rows are exact, and second rows are approximations (3.4) and (3.5).

α/β	0.0	0.1	0.2	0.3	0.4	0.5	0.6
$\Delta E_s/D$							
0.005	1.000	0.332	0.106	0.0238	0.00378	0.000312	8.4×10^{-6}
		0.357	0.105	0.0238	0.00381	0.000374	1.8×10^{-5}
0.01	1.000	0.358	0.120	0.0321	0.00600	0.000625	2.3×10^{-5}
		0.388	0.126	0.0321	0.00594	0.000701	4.3×10^{-5}
0.02	1.000	0.387	0.142	0.0431	0.00952	0.00125	6.6×10^{-5}
		0.421	0.150	0.0431	0.00924	0.00131	1.0×10^{-4}
0.04	1.000	0.418	0.169	0.0581	0.0151	0.00250	1.9×10^{-4}
		0.457	0.180	0.0581	0.0144	0.00246	2.4×10^{-4}
0.08	1.000	0.451	0.201	0.0782	0.0240	0.00500	5.3×10^{-4}
		0.496	0.215	0.0781	0.0224	0.00460	5.8×10^{-4}

$$C_1 = 2.962 - 1.086 \ln(\Delta E_s/D). \quad (3.5)$$

The agreement between the empirical representation and the exact calculation is satisfactory.

After this, for a nonlinear triatomic molecule we calculate the threshold energy pattern ϵ_i^* of a single

disappearing oscillator (energy quantum ϵ_e) which is transformed into a product rotor. For the nonrotating molecule ($J=0$), the product rotation is nondegenerate, the correlation scheme identifies the quantum numbers $i \equiv v=l=j$, and one has to determine the maxima of the channel potential

$$V_a(q) = D[1 - \exp[-\beta(q - q_e)]]^2 + (v+1/2)\epsilon_e \exp[-\alpha(q - q_e)] + j(j+1)B_\infty\{1 - \exp[-\alpha(q - q_e)]\} + B_{\text{cent}}(q)P(P+1), \quad (3.6)$$

with

$$P = l\{1 - \exp[-\alpha(q - q_e)]\} \quad (3.7)$$

and

$$B_{\text{cent}}(q) \approx B_e \frac{m_A(m_B + m_C)q_{2e}^2 - 2m_A m_C q_{2e} q_e \cos \vartheta_e + m_C(m_A + m_B)q_e^2}{m_A(m_B + m_C)q_{2e}^2 - 2m_A m_C q_{2e} q_e \cos \vartheta_e + m_C(m_A + m_B)q_e^2}. \quad (3.8)$$

The centrifugal energy for the orbital motion to a good approximation can be expressed by the smallest rotational constant only [Eq. (3.8)] since with increasing reaction coordinate q the difference between the two smallest rotational constants rapidly decreases. In Eq. (3.8), the nonlinear molecule ABC is characterized by the bond angle ϑ_e , the equilibrium bond length q_{2e} of the A-BC bond, and the equilibrium value q_e of the length of the AB-C bond q which is identified with the reaction coordinate. m_A , m_B , and m_C are the respective masses of A, B, and C. B_e (in energy units) is taken as the mean $(B+C)/2$ of the two smallest rotational constants of the reactant molecule in the equilibrium configuration. Abbreviating $\beta(q - q_e) = z$, the centrifugal energy can also be written

$$B_{\text{cent}}(z) = B_e/(1 + a_1 z + a_2 z^2), \quad (3.9)$$

with the parameters

$$a_1 = \frac{2}{\beta q_e} \left[\frac{-m_A m_C q_{2e} q_e \cos \vartheta_e + m_C(m_A + m_B)q_e^2}{m_A(m_B + m_C)q_{2e}^2 - 2m_A m_C q_{2e} q_e \cos \vartheta_e + m_C(m_A + m_B)q_e^2} \right] \quad (3.10)$$

and

$$a_2 = \left(\frac{1}{\beta q_e} \right)^2 \left[\frac{m_C(m_A + m_B)q_e^2}{m_A(m_B + m_C)q_{2e}^2 - 2m_A m_C q_{2e} q_e \cos \vartheta_e + m_C(m_A + m_B)q_e^2} \right]. \quad (3.11)$$

For a quasidiatomic molecule, $a_1 = 2/\beta q_e$ and $a_2 = 1/(\beta q_e)^2$.

The channel maxima in Eq. (3.6) depend on the parameters a_1 , a_2 , B_e/D , B_∞/D , ϵ_e/D , and α/β . In the following, the maxima of $V_a(q)$ for varying q , for a given $v=j=l=i$, are identified with $E_0 + \epsilon_i^*$. [In this treatment, for simplicity, $\epsilon_e/2$ is identified with ΔE_s and the constant term E_{zp} is omitted. Whereas this simplification cannot be made in the calculation of the adiabatic

zero point barriers (3.2)–(3.4), the threshold pattern ϵ_i^* for $i > 0$ is only little influenced.] Since ϵ_i^* increases monotonically with increasing channel quantum number i , the relation between ϵ_i^* and i directly corresponds to the number of open channels $W^*(E, J=0)$ as a function of the energy E [see Eq. (1.3)]: One simply has to identify $i+1$ with W^* and ϵ_i^* with E .

Up to this point, we have followed the statistical

adiabatic channel model of Ref. 4. We now analyze the calculated threshold energy pattern in Eq. (3.6). We express it empirically by a semiclassical expression

$$W^* \approx \left(\frac{E-D}{\epsilon} \right)^x, \quad (3.12)$$

where the two parameters ϵ and x depend on a_1 , a_2 , B_e/D , B_∞/D , ϵ_e/D , and α/β . Since the correct W^* is a step function, its "translation" into the continuous function (3.12) and its requantization have to be specified consistently. In the present work, we connect all steps at one half their height to obtain Eq. (3.12). Requantization of the semiclassical number of channels W_{sc}^* is performed by the relation

$$W^* = \text{Int}(W_{sc}^* + 0.5), \quad (3.13)$$

where $\text{Int}(W_{sc}^* + 0.5)$ denotes the next nearest integer below $W_{sc}^* + 0.5$. Requantization will be particularly important for the calculation of specific rate constants $k(E, J)$.²⁴ Equation (3.12), for a harmonic oscillator energy pattern of ϵ_i^* , i.e.,

$$\epsilon_i^* - D = (i + \frac{1}{2})\epsilon_e, \quad (3.14)$$

corresponds to

$$W^* = \left(\frac{E-D}{\epsilon_e} \right). \quad (3.15)$$

For a nondegenerate one-dimensional rotor pattern

$$\epsilon_i^* - D = i(i+1)B_\infty, \quad (3.16)$$

Eq. (3.12) corresponds to

$$W^* \approx \left(\frac{E-D}{B_\infty} \right)^{0.5}. \quad (3.17)$$

The true threshold energy pattern will be embraced by the two limits (3.15) and (3.17). Expression (3.12) will be of use whenever a constant value of the exponent x arises which is independent of the energy E .

Figures 1–3 show the calculated channel maxima patterns from Eq. (3.6) as a function of the parameters α/β for $\epsilon_e/D=0.005$, 0.02, and 0.08, respectively, with $B_\infty/D=2 \times 10^{-5}$ and $B_e/D=0$. The linear relationships in

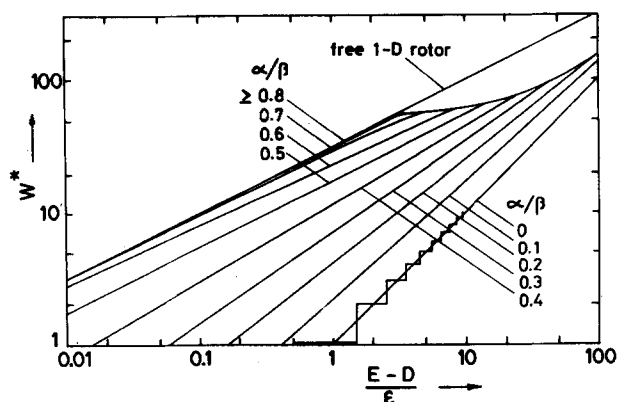


FIG. 1. Number of open channels W^* as a function of the energy E for a single channel ($\epsilon_e/D=0.02$, $B_\infty/D=2 \times 10^{-5}$, $B_e/D=0$; identical results, for example, for $B_e/D=10^{-5}$; $a_1=0.3$, $a_2=0.1$).

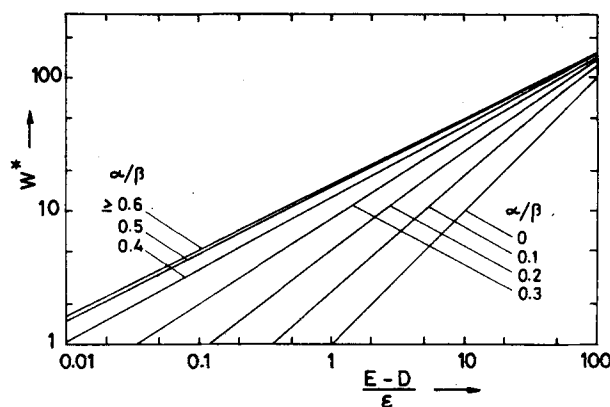


FIG. 2. The same as Fig. 1, but $\epsilon_e/D=0.005$.

the double logarithmic plot over a wide range of conditions show indeed that Eq. (3.12) with energy independent x gives a useful representation. Deviations from this behavior only occur at high energies, when centrifugal effects of the orbital motion turn the channel potential into purely repulsive curves, and for $\alpha/\beta \rightarrow 1$ when nearly free rotors appear. These ranges are only of marginal practical importance and are neglected in the following. Inspecting systematically the dependence of x and ϵ in Eq. (3.12) on the parameters given above showed primarily a dependence on α/β . An additional, much smaller dependence on ϵ_e/D and ϵ_e/B_∞ was noticed. The influence of variations of B_e/D in the range $0-(6 \times 10^{-4})$ for $a_1 \leq 1$ and $a_2 \leq 1$ was negligibly small. The dependence of x and ϵ on these parameters can be expressed conveniently in an empirical way. By fixing x via the W^* values at 10 and 20, we obtained results like those shown in Table III. They are well reproduced by the relation

$$x = x_e \exp(-y - y^n) + x_\infty [1 - \exp(-y - y^n)]^n, \quad (3.18)$$

with $x_e=1$, $x_\infty=0.5$, and

$$y = C_2 \alpha/\beta, \quad (3.19)$$

$$C_2 \approx 1.24 + 55 B_\infty/\epsilon_e, \quad (3.20)$$

$$n \approx 2.25 + (0.5 \times 10^{-3}) \epsilon_e/B_\infty. \quad (3.21)$$

In the same way, from the values of W^* at 10 and 20, we obtained values for ϵ . Expressing these by the relation

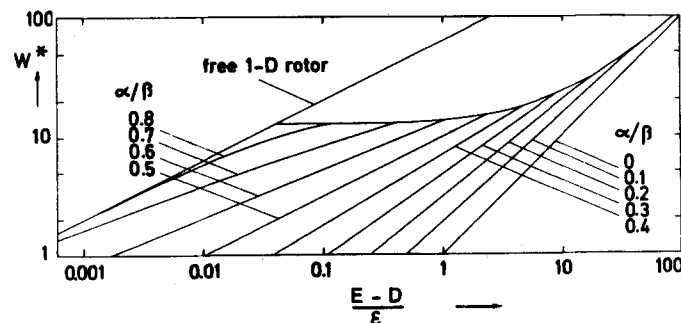


FIG. 3. The same as Fig. 1, but $\epsilon_e/D=0.08$.

TABLE III. Exponents x in the number of open channels W^* [Eq. (3.12)]. Upper half: $B_\infty/D = 5 \times 10^{-6}$; lower half: $B_\infty/D = 2 \times 10^{-5}$. First rows are exact; second rows are from Eqs. (3.18)–(3.21).

$\epsilon_g/D \backslash \alpha/\beta$	0.0	0.2	0.3	0.4	0.5	0.6	0.7	0.8	1.0
0.005	1.00	0.78	0.67	0.57	0.50	0.49	0.50	0.50	0.50
	1.00	0.76	0.66	0.58	0.52	0.48	0.47	0.47	0.49
0.01	1.00	0.79	0.69	0.59	0.50	0.46	0.49	0.50	0.50
	1.00	0.77	0.67	0.58	0.51	0.46	0.44	0.45	0.48
0.02	1.00	0.80	0.70	0.60	0.50	0.42	0.42	0.49	0.50
	1.00	0.78	0.68	0.59	0.50	0.43	0.40	0.41	0.48
0.04	1.00	0.80	0.69	0.59	0.50	0.40	0.33	0.37	0.50
	1.00	0.78	0.69	0.60	0.51	0.42	0.34	0.34	0.49
0.005	1.00	0.74	0.63	0.54	0.50	0.50	0.50	0.50	0.50
	1.00	0.73	0.63	0.56	0.51	0.49	0.49	0.49	0.50
0.01	1.00	0.78	0.67	0.57	0.50	0.49	0.50	0.50	0.50
	1.00	0.75	0.65	0.57	0.52	0.49	0.48	0.48	0.49
0.02	1.00	0.79	0.67	0.59	0.50	0.46	0.47	0.50	0.50
	1.00	0.76	0.66	0.58	0.52	0.48	0.47	0.47	0.49
0.04	1.00	0.79	0.69	0.59	0.50	0.42	0.38	0.45	0.50
	1.00	0.77	0.67	0.58	0.51	0.46	0.44	0.45	0.48

$$\epsilon \cong B_\infty + (\epsilon_g - B_\infty) \exp[-C_3 \alpha/\beta - C_4 (\alpha/\beta)^4] \quad (3.22)$$

$$C_4 \cong 2.80 - 5.19 \ln(\epsilon_g/D) \quad (3.24)$$

gives the values of $(\epsilon - B_\infty)/(\epsilon_g - B_\infty)$ which are compared in Table IV with the numerical results. The parameters C_3 and C_4 are represented here by

$$C_3 \cong 4.00 - 1.29 \ln(\epsilon_g/D) \quad (3.23)$$

and

As in Table III, the agreement between the exact and the empirically approximated values is quite good.

So far, the energy pattern of channel maxima has been considered only for excitation of individual coordinates. In reality, however, there will be a consider-

TABLE IV. Energy quanta ϵ , expressed by $(\epsilon - B_\infty)/(\epsilon_g - B_\infty)$, in the number of open channels W^* [Eq. (3.12)]. Upper half: $B_\infty/D = 5 \times 10^{-6}$; lower half: $B_\infty/D = 2 \times 10^{-5}$. First rows are exact; second rows are from Eqs. (3.22)–(3.24).

$\epsilon_g/D \backslash \alpha/\beta$	0.0	0.2	0.3	0.4	0.5	0.6	0.7	0.8	1.0
0.005	1.00	0.12	0.030	0.0057	0.00064	-2.1×10^{-5}	-1.7×10^{-5}	-9.4×10^{-6}	-9.5×10^{-6}
	1.00	0.11	0.030	0.0060	0.00067	$+3.0 \times 10^{-5}$	$+3.5 \times 10^{-7}$	$+7.0 \times 10^{-10}$	$+1.4 \times 10^{-8}$
0.01	1.00	0.14	0.042	0.0094	0.0013	-1.1×10^{-5}	-4.4×10^{-5}	-5.6×10^{-6}	-4.7×10^{-6}
	1.00	0.13	0.041	0.0095	0.0013	$+8.1 \times 10^{-5}$	$+1.6 \times 10^{-6}$	$+6.3 \times 10^{-9}$	$+1.2 \times 10^{-16}$
0.02	1.00	0.17	0.057	0.015	0.0025	0.00011	-1.1×10^{-4}	-1.7×10^{-5}	-2.4×10^{-6}
	1.00	0.16	0.055	0.015	0.0026	0.00022	$+6.9 \times 10^{-6}$	$+5.6 \times 10^{-8}$	$+1.1 \times 10^{-14}$
0.04	1.00	0.20	0.077	0.023	0.0048	0.00046	-8.7×10^{-5}	-9.6×10^{-5}	-1.2×10^{-6}
	1.00	0.19	0.074	0.023	0.0050	0.00060	$+3.1 \times 10^{-5}$	$+5.0 \times 10^{-7}$	$+9.7 \times 10^{-13}$
0.005	1.00	0.11	0.027	0.0054	0.00069	-3.0×10^{-5}	-4.3×10^{-5}	-3.8×10^{-5}	-3.8×10^{-5}
	1.00	0.11	0.030	0.0060	0.00067	$+3.0 \times 10^{-5}$	$+3.5 \times 10^{-7}$	$+7.0 \times 10^{-10}$	$+1.4 \times 10^{-18}$
0.01	1.00	0.14	0.040	0.0091	0.0013	-2.9×10^{-5}	-5.5×10^{-5}	-2.0×10^{-5}	-1.9×10^{-5}
	1.00	0.13	0.041	0.0095	0.0013	$+8.1 \times 10^{-5}$	$+1.6 \times 10^{-6}$	$+6.3 \times 10^{-6}$	$+1.2 \times 10^{-16}$
0.02	1.00	0.17	0.057	0.015	0.0026	1.8×10^{-5}	-1.8×10^{-4}	-2.4×10^{-5}	-9.4×10^{-6}
	1.00	0.16	0.055	0.015	0.0026	2.2×10^{-4}	$+6.9 \times 10^{-6}$	$+5.6 \times 10^{-8}$	$+1.1 \times 10^{-14}$
0.04	1.00	0.20	0.076	0.023	0.0049	0.00035	-3.1×10^{-4}	-1.8×10^{-4}	-4.7×10^{-6}
	1.00	0.19	0.074	0.023	0.0050	0.00060	$+3.1 \times 10^{-5}$	$+5.0 \times 10^{-7}$	$+9.7 \times 10^{-13}$

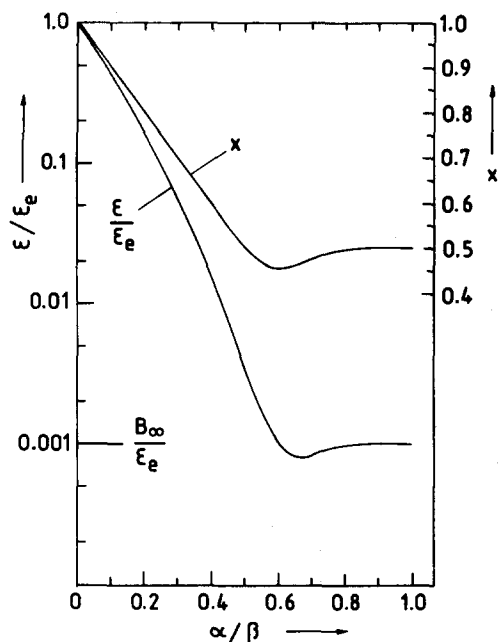


FIG. 4. Effective quanta ϵ and exponents x in the number of open channels W^* [Eq. (3.12)]. (For $\epsilon_e/D=0.02$, $B_\infty/D=2 \times 10^{-5}$, $B_0/D=0$.)

able influence of the excitation of some coordinates on the channel pattern of other coordinates. We have to treat this effect in an approximate way by considering the channel potential (2.1) for excitation of several coordinates:

$$V_a(q) = V(q) + E_{ar}(q) + E_{ab}(q) + E_{\text{cent}}(q) + E_{at}(q) \\ = V_{\text{eff}}(q) + E_{at}(q). \quad (3.25)$$

Here $E_{ar}(q)$ denotes the channel energies of conserved oscillators (and rotors), $E_{ab}(q)$ is the channel energy for disappearing oscillators different from the oscillator under consideration whose channel energy is denoted by $E_{at}(q)$, and $E_{\text{cent}}(q)$ is the centrifugal energy. The term $E_{ar}(q)$ does not vary markedly with q , whereas $E_{ab}(q)$ and $E_{\text{cent}}(q)$ according to Eqs. (2.5)–(2.7) depend strongly on q . For not too high excitations of the fragment states, $E_{ab}(q)$ drops from oscillator states into rotor states such that the effective potential $V_{\text{eff}}(q)$ in Eq. (3.25) appears to have a smaller dissociation energy than the Morse potential. This effect is further increased by the contribution from $E_{\text{cent}}(q)$. Correspondingly [Eq. (2.4)], there is an increase of the effective Morse parameter β . (The corresponding effect, due to variations in the force constant, is always negligible.) In the present model, we treat this effect by approximately putting β_{eff} instead of β :

$$\beta_{\text{eff}} \approx \beta \sqrt{\frac{D}{D - b' \langle E \rangle / s - \langle E_{\text{cent}}(q_e) \rangle}}, \quad (3.26)$$

where $b' = b - 1$ for a linear molecule (or $b' = b - 2$ for a nonlinear molecule), s is the number of oscillators, $\langle E \rangle$ is the average energy of the molecule in its reacting states, and $\langle E_{\text{cent}}(q_e) \rangle$ is the average centrifugal energy at $q = q_e$. For a thermal situation, $\langle E_{\text{cent}}(q_e) \rangle$ is given by kT , and $\langle E \rangle$ is equal to $D + \langle U_{\text{vib}} \rangle = D + s_{\text{eff}} kT$ (rough

estimates of s_{eff} are required only). Equation (3.26) attributes an average excitation of $\langle E \rangle / s$ to the oscillators. In this way, it accounts for the difference between the Morse potentials $V(q)$ and $V_{\text{eff}}(q)$ in an averaged manner. $\alpha/\beta_{\text{eff}}$ then replaces α/β in Eqs. (3.19) and (3.22). Similarly, $D - b' \langle E \rangle / s - \langle E_{\text{cent}}(q_e) \rangle$ replaces D in Eqs. (3.23) and (3.24).

The quality of the relationships (3.18)–(3.24) and (3.26), and of the simplifications made so far, can only be tested by comparison with the detailed statistical adiabatic channel model for practical systems (see below). For better accuracy, the detailed model should be taken.

Figures 1–4 illustrate a number of most interesting properties of the channel threshold energy pattern in the present model. The disappearing oscillators have a threshold pattern which neither corresponds to states of harmonic oscillators nor of free or hindered rotors. This threshold energy pattern, therefore, allows to calculate pseudopartition functions Q^* of activated complexes.

IV. THRESHOLD ENERGY PATTERN FOR CONSERVED OSCILLATORS AND ROTORS

Oscillators, which are conserved in the reaction, show a channel threshold pattern ϵ_{ji} in Eqs. (3.1) which differs from the expression (3.12). In contrast to the generally very large difference between ϵ_e and B_∞ , here the channel eigenvalues $E_a(q)$ do not vary too much with q such that, for the considered case of a simple bond fission, the channel threshold energies ϵ_{ji} are very close to the energy states of the corresponding product oscillators. For this reason, in our model a one-to-one correlation between reactant and product oscillators is not required.

If among the “conserved coordinates” a reactant internal rotor is transformed into a product oscillator, or *vice versa*, for simplicity the energy pattern of the product states can be used for ϵ_{ji} . If one wishes to do better, or if the full transition between the rigid limit $\alpha/\beta=0$ and the loose limit $\alpha/\beta \gtrsim 1$ is to be explored, the conserved coordinates can be treated analogously by to Sec. III, i.e., in the simplest way, x and ϵ are interpolated by

$$x \approx x_\infty + (x_e - x_\infty) \exp(-C_2 \alpha/\beta), \quad (4.1)$$

$$\epsilon \approx \epsilon_\infty + (\epsilon_e - \epsilon_\infty) \exp(-C_3 \alpha/\beta), \quad (4.2)$$

where

$$C_2 \approx 1.24 + 55 \epsilon_\infty / \epsilon_e, \quad (4.3)$$

$$C_3 \approx 4.00 - 1.29 \ln(\epsilon_e/D). \quad (4.4)$$

These equations are given without justification. They are of minor importance, and they only assure that the conserved coordinates are interpolated between the correct rigid and loose limits in a way which is consistent with the results of Sec. III. For a nonlinear reactant molecule, one external rotor (quantum number τ_n for an asymmetric top, or quantum number K for a prolate symmetric top with largest rotational constant A_e) is transformed in one of the product rotors (see above).

In this case, x is equal to 0.5, and ϵ , similar to the case of conserved oscillators, is given by a rotational constant which via Eq. (3.22) is interpolated between A_e and B_∞ .

The summation in Eq. (1.2) includes all channel states. If a "mechanical counting" is performed neglecting symmetry exclusion rules, the resulting numbers of channels W^* and partition functions Q_m and Q_m^* have to be divided by the relevant symmetry numbers σ and σ^* , respectively. To a first approximation, σ^* is interpolated between the rigid and loose limits similar to Eq. (4.2) by

$$\sigma^* \approx \sigma_1 \sigma_2 S_\infty + (\sigma S_e - \sigma_1 \sigma_2 S_\infty) \exp(-C_3 \alpha / \beta), \quad (4.5)$$

where σ_1 and σ_2 are the symmetry numbers of the two fragment molecules. $S_\infty = 2$ if the fragments are identical, or $S_\infty = 1$ otherwise. As suggested in Ref. 17, $S_e \approx \sigma_e / \sigma$ is in general adequate with σ_e being taken as the symmetry number of the reactant molecule when it is distorted along the reaction coordinate q , but otherwise kept in the most symmetrical conformation.

V. CENTRIFUGAL BARRIER PATTERN

Having represented the threshold energy patterns ϵ_{ji} and ϵ_{ji}^* of conserved and disappearing oscillators and rotors in Eq. (3.1) by empirical relationships, we now derive the corresponding expressions for the centrifugal barriers $E_0(J)$. The quantities $E_0(J)$ are given by the maxima of the lowest channel potential for a given J , i.e., of

$$V_a(q) = D \{1 - \exp[-\beta(q - q_e)]\}^2 + \Delta E_e \exp[-\alpha(q - q_e)] + E_{zp} + B_{\text{cent}}(q) J(J+1) \quad (5.1)$$

[see Eqs. (2.5)–(2.7), (3.2), and (3.6–3.11) with $J = l = P$]. For the present purposes, an exact calculation of $B_{\text{cent}}(q)$ hardly is required. Instead, polyatomic molecules are represented either as quasidiatomic or quasitriatomic molecules, such that $B_{\text{cent}}(q)$ is given by Eqs. (3.9)–(3.11), i.e.,

$$V_a(z) = D \{1 - \exp(-z)\}^2 + \Delta E_e \exp\left(-\frac{\alpha}{\beta} z\right) + E_{zp} + B_e J(J+1) \left(\frac{1}{1 + a_1 z + a_2 z^2} \right). \quad (5.2)$$

According to Eq. (5.2), the heights of the centrifugal barriers depend on the parameters α/β , $\Delta E_e/D$, B_e/D , a_1 , and a_2 . They are easily obtained by numerical determination of the maximum of $V_a(z)$ with varying z . The inspection of Eq. (5.2), as a general result, shows that centrifugal barriers can be represented in the form³

$$E_0(J) - E_0(J=0) \approx C_\nu [J(J+1)]^\nu, \quad (5.3)$$

where the exponents ν range between 1 and 1.5. Therefore, centrifugal maxima for two J values allow a determination of ν and C_ν . Figure 5 illustrates the centrifugal barrier relation by a $\log[E_0(J) - E_0(J=0)] - \log[J(J+1)]$ plot for a typical example.

In practice, the parameters ν and C_ν sensitively enter into the rate constant. Therefore, small deviations from Eq. (5.3) can have some effect. In order to cor-

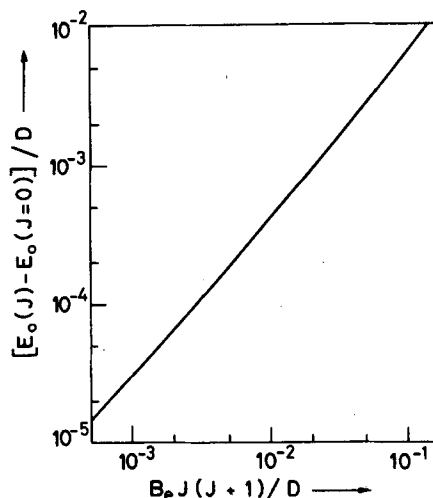


FIG. 5. Centrifugal barriers from Eq. (5.2). (With $\Delta E_e/D = 0.01$, $\alpha/\beta = 0.6$, $a_1 = 0.7$, and $a_2 = 0.13$.)

rect for this, we propose the following simple procedure: At first, for a conveniently small and a large value, $E_0(J) - E_0(J=0)$ is determined, and a first guess for C_ν and ν is derived from Eq. (5.3). In the centrifugal pseudopartition function Q_{cent}^* (see Sec. VI) one sums over $f(J) = (2J+1) \exp\{-C_\nu [J(J+1)]^\nu / kT\}$. Here, the main contributions to Q_{cent}^* stem from J values between 0 and $2J_{\text{max}}(T)$, where $J_{\text{max}}(T)$ corresponds to the maximum of the distribution $f(J)$ given by

$$J_{\text{max}}(T) \approx \left(\frac{kT}{2\nu C_\nu} \right)^{1/2\nu} - \frac{1}{2}. \quad (5.4)$$

Entering with the first guess values of ν and C_ν into Eq. (5.4) gives $J_{\text{max}}(T)$ and allows for a determination of effective, slightly temperature dependent values of ν and C_ν from the centrifugal maxima at a conveniently small J value and at $J \approx 2J_{\text{max}}(T)$. An illustration for such a calculation is given in Sec. VIII. With a computer, of course, the exact centrifugal barrier pattern could also be calculated from Eq. (5.2).

The centrifugal maxima in general are influenced by the potential $\Delta E_e \exp[-(\alpha/\beta)z]$ of the lowest channel state as indicated by Eqs. (5.1) and (5.2). In loose activated complex theory ($\alpha/\beta \geq 1$), this influence disappears. In order to simplify the calculation of the centrifugal barrier parameters ν and C_ν of Eq. (5.3), at least for this limiting case, we have systematically evaluated centrifugal barriers by numerical calculations. For rotational energies $B_e J(J+1)/D$ between 0.005 and 0.05 we have derived an empirical representation of ν and C_ν as a function of the parameters a_1 and a_2 . With the abbreviations

$$V = 4a_2/a_1^2, \quad (5.5)$$

$$C_5 = 0.61 - 0.20 \exp(-0.256 V), \quad (5.6)$$

$$C_6 = 11.7 - 8.6 \exp(-0.232 V), \quad (5.7)$$

$$C_7 = 1 + 0.25 V, \quad (5.8)$$

$$C_8 = 2 + 0.66 V, \quad (5.9)$$

$$C_9 = 16.6 + 0.25 V, \quad (5.10)$$

$$Y = C_7 \exp(-C_8 \alpha_1) - (C_7 - 1) \exp(C_9 \alpha_1), \quad (5.11)$$

we obtained

$$\nu \approx 1 + C_5 [1 - \exp(-C_6 \alpha_1)], \quad (5.12)$$

$$C_\nu \approx Y D (B_e/D)^\nu. \quad (5.13)$$

With these parameters, centrifugal barrier partition functions are directly obtained as shown in the next section. These partition functions, if evaluated with Eqs. (5.5)–(5.13), for the loose limit are generally accurate within 10%. The empirical representations of ν and C_ν in the general case $\alpha/\beta < 1$ have a much more complicated form such that in this case Eq. (5.2) should rather be maximized directly as described above.

The calculation of ν and C_ν give results which are considerably different from those obtained for a van der Waals potential $V(q) = -C^*/q^6$ and quasidiatomic centrifugal potential, for which³ $C_\nu = \sqrt{4B_e^3 q_0^3 / 27C^*}$ and $\nu = 1.5$. In addition, one notices a marked influence of the adiabatic zero point barriers (via the parameters $\Delta E_e/D$ and α/β) on the centrifugal barrier pattern. The appearance of these adiabatic zero point barriers leads to lower values of ν approaching 1 and to higher centrifugal barriers, i.e., higher values of C_ν .

VI. PSEUDOPARTITION FUNCTION OF THE ACTIVATED COMPLEX

The pseudopartition function Q^* can be calculated for separated conserved and disappearing oscillators with the channel pattern analyzed in Secs. III and IV without too much difficulty, if angular momentum coupling restrictions, couplings between various coordinates, and dependences of their state pattern on J are neglected. Angular momentum coupling can be corrected for at the end. Other couplings cannot be treated in this simple way, but can only be considered in the detailed statistical adiabatic channel model.⁴

Splitting Eq. (1.2) into factors, one obtains with Eq. (3.1)

$$Q^* = \frac{1}{\sigma^*} F_{AM}^* Q_{cent}^* \prod_{j=1}^r Q_j^* \prod_{m=1}^b Q_m^*, \quad (6.1)$$

where σ^* is the effective symmetry number (4.5), Q_{cent}^* is the centrifugal pseudopartition function

$$Q_{cent}^* = \sum_{J=0}^{\infty} (2J+1) \exp\left[-\frac{E_0(J) - E_0}{kT}\right], \quad (6.2)$$

where E_0 as always is identical to $E_0(J=0)$, Q_j^* are the partition functions of the r conserved oscillators, Q_m^* are the partition functions of the b disappearing oscillators and rotors, and F_{AM}^* is the angular momentum coupling correction (see below). If there are nonnegligible adiabatic zero point barriers, E_0 corresponds to $\Delta H_0^0 + \Delta E_{0e}$ [see Eq. (3.4)].

Replacing the summation over J by an integration, the centrifugal pseudopartition function (6.2) can directly be evaluated³:

$$Q_{cent}^* \approx \Gamma\left(1 + \frac{1}{\nu}\right) \left(\frac{kT}{C_\nu}\right)^{1/\nu}. \quad (6.3)$$

The partition functions Q_j of the conserved oscillators

are quantum partition functions

$$Q_j^* = \frac{1}{1 - \exp\left(-\frac{\epsilon_j}{kT}\right)}. \quad (6.4)$$

Since the quantities ϵ_m for the disappearing oscillators and rotors, for typical α/β values, are in most cases much smaller than kT , classical partition functions can be employed:

$$Q_m^* \approx \int_0^\infty \frac{dW_m^*(E)}{dE} \exp\left(-\frac{E}{kT}\right) dE = \Gamma(1 + x_m) \left(\frac{kT}{\epsilon_m}\right)^{x_m}. \quad (6.5)$$

Here, W_m^* from Eq. (3.12) has been used. If, in the general case, quantization for the disappearing oscillators is required, Eq. (6.5) is replaced by the pseudopartition function

$$Q_m^* \approx \Gamma(1 + x_m) \left[\frac{1}{1 - \exp\left(-\frac{\epsilon_m}{kT}\right)} \right]^{x_m}. \quad (6.6)$$

Equation (6.6) includes the limits of a harmonic oscillator and a one-dimensional classical rotor.

The factorization and separation of the loosening oscillators and rotors, because of the neglect of angular momentum constants, has introduced errors which we correct *a posteriori* by the factor F_{AM}^* . If one had to consider only harmonic oscillators, no such correction would be required. It is the "rotational character" of the coordinates which is responsible for the correction. It, therefore, appears plausible to relate F_{AM}^* to the extent of rotational character of the disappearing oscillator. In the threshold energy pattern analyzed in Sec. III, this rotational character manifests itself in the interpolation laws for the quanta ϵ [Eq. (3.22)] and exponents x [Eq. (3.18)]. We propose, therefore, to relate the rotational character to the quanta ϵ by interpolating F_{AM}^* between an oscillator and a rotor situation in a simple way analogous to Eq. (3.22):

$$F_{AM}^* \approx F_{AM_\infty}^* + (F_{AM_e}^* - F_{AM_\infty}^*) \exp(-C_3 \alpha/\beta), \quad (6.7)$$

with C_3 given by Eq. (3.23).

The corrections $F_{AM_e}^*$ and $F_{AM_\infty}^*$ correspond to the limits of rigid ($\alpha/\beta=0$) and loose ($\alpha/\beta=\infty$) activated complex theory. They are easily obtained by comparing the factorized Q^* with the correct expression. For instance, in the loose limit, for two nonlinear fragments the correct partition function (neglecting symmetry) of the oscillators would be $\sqrt{\pi(kT)^3/A_1 B_1 C_1} \sqrt{\pi(kT)^3/A_2 B_2 C_2}$, whereas ΠQ_m^* with Eq. (6.5) would give $(\sqrt{\pi}/2)^6 \times \sqrt{(kT)^3/A_1 B_1 C_1} \sqrt{(kT)^3/A_2 B_2 C_2}$; therefore, in this case, $F_{AM_\infty}^* = 2^6/\pi^2$. $F_{AM_\infty}^*$ for other cases are given in Table V. For the rigid limit, $F_{AM_e}^*$ is equal to unity for a linear reactant molecule; for a nonlinear reactant, $F_{AM_e}^* \approx 2$ is good approximation (see Appendix A). Since Eq. (6.7) refers to some overall rotational character as expressed by Eq. (3.22), the quantity ϵ_e/D in the expression (3.23) for C_3 should be the average over all disappearing oscillators (rotors not included). [Similarly, the symmetry number in Eq. (4.5) should be interpolated with an average ϵ_e/D .] There is no justification for the angular momentum correction (6.7) other than the fact that it inter-

TABLE V. Angular momentum coupling corrections F_{AM}^* for the rigid (F_{AM}^*) and loose limit ($F_{AM\infty}^*$).

Products	$F_{AM\infty}^*$
Nonlinear + nonlinear	$2^6/\pi^2$
Nonlinear + linear	$2^5/\pi^2$
Nonlinear + atom	$2^3/\pi$
Linear + linear	$2^4/\pi^2$
Linear + atom	$2^2/\pi$
Reactant	F_{AM}^*
Linear	1
Nonlinear	$\approx 2^2$

^aSee Appendix A.

polates between the correct expressions for the accessible limits of rigid and loose transition state theory in a plausible and simple way. Therefore, a comparison of the present simplified statistical adiabatic channel model with the detailed full version of the theory is required. This will be given for a number of practical examples in the last section.

VII. APPLICATIONS AND COMPARISONS

In the following we apply the present theory in detail to a number of practical examples so that users have a guideline. We also compare our results with full statistical adiabatic channel calculations from earlier work. For a comparison, we can either choose the high pressure dissociation rate constant

$$k_{\infty} = \frac{kT}{h} \frac{Q_{\text{cent}}^* F_{AM}^* \frac{1}{\sigma^*} \prod_j Q_j^* \prod_m Q_m^*}{Q_{\text{vibrot}}(A)} \times \exp\left(-\frac{\Delta E_{0g}}{kT}\right) \exp\left(-\frac{\Delta H_0^{\ddagger}}{kT}\right) \quad (7.1)$$

or the high pressure recombination rate constant

$$k_{\text{rec},\infty} = \frac{kT}{h} \left(\frac{h^2}{2\pi\mu kT}\right)^{3/2} \frac{Q_{e1}(A)}{Q_{e1}(B)Q_{e1}(C)} \times \frac{Q_{\text{cent}}^* F_{AM}^* \frac{1}{\sigma^*} \prod_j Q_j^* \prod_m Q_m^*}{Q_{\text{vibrot}}(B)Q_{\text{vibrot}}(C)} \exp\left(-\frac{\Delta E_{0g}}{kT}\right) \quad (7.2)$$

or the corresponding high pressure recombination cross section

$$\sigma_{\text{rec},\infty} = k_{\text{rec},\infty} / \langle v \rangle. \quad (7.3)$$

Here, Q_{vibrot} are rovibrational partition functions including symmetry corrections, Q_{e1} are electronic partition functions, μ is the reduced mass $\mu = m_B m_C / m_A$ for the reaction $A \rightarrow B + C$, and $\langle v \rangle$ is the average velocity $\langle v \rangle = (8kT/\pi\mu)^{1/2}$.

Often the use of $k_{\text{rec},\infty}$ is most convenient, since Eq. (7.2) can be further simplified. For $0.01 \leq \epsilon_g/D \leq 0.1$ in Eq. (3.23) one has $7 \leq C_3 \leq 10$, such that for $\alpha/\beta \geq 0.3$ one obtains $\exp(-C_3\alpha/\beta) < 0.12$. Therefore, the effective quanta for conserved oscillators are practically given by the corresponding quanta for fragment vibrations (only when $\alpha/\beta < 0.3$, or when internal rotors are transformed into oscillators and *vice versa*, is a more detailed treatment following Sec. IV required). As a consequence, the fragment vibrational partition functions in Eq. (7.2) cancel against $\prod_j Q_j^*$ such that

$$k_{\text{rec},\infty} \cong \frac{kT}{h} \left(\frac{h^2}{2\pi\mu kT}\right)^{3/2} \frac{Q_{e1}(A)}{Q_{e1}(B)Q_{e1}(C)} \times \frac{Q_{\text{cent}}^* F_{AM}^* \frac{1}{\sigma^*} \prod_m Q_m^*}{Q_{\text{rot}}(B)Q_{\text{rot}}(C)} \exp\left(-\frac{\Delta E_{0g}}{kT}\right). \quad (7.4)$$

We evaluate Eq. (7.4) for the systems $\text{O}_3 \rightleftharpoons \text{O} + \text{O}_2$, $\text{NO}_2 \rightleftharpoons \text{O} + \text{NO}$, $\text{ClNO} \rightleftharpoons \text{Cl} + \text{NO}$, and $\text{C}_2\text{H}_6 \rightleftharpoons 2\text{CH}_3$, which were treated by the statistical adiabatic channel model in Refs. 4, 5, and 17. The results are shown in Table VI and compared with the values obtained for the same values of α and β from the complete statistical adiabatic channel model (Refs. 4 and 5), and from a maximum free energy calculation with exponentially interpolated free energy (with $\gamma = 0.75\alpha$, see Ref. 17).

TABLE VI. High pressure recombination rate constant (Exp: experimental data, references in Table I of Ref. 17; ACM: statistical adiabatic channel calculations of Refs. 4 and 5, chosen α values in parentheses; Max A: maximum free energy calculation of Ref. 17; this work: Eq. (7.4); details for C_2H_6 , see Appendix B; all molecular parameters are as given in Refs. 4, 5, and 17).

$k_{\text{rec},\infty} (10^{-11} \text{ cm}^3 \text{ s}^{-1})$						
Reaction	T (K)	Exp	ACM ($\alpha/\text{\AA}^{-1}$) ^{4,5}	Max A ¹⁷ with $\gamma = 0.75\alpha$	This work: α as ACM	This work: phase space limit: $\alpha/\beta \geq 1$
$\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$	300	4.8	5.1 (1.0)	4.65	5.5	18
	1300	2.9	4.8 (1.0)	6.23	5.7	24
$\text{O} + \text{NO} \rightarrow \text{NO}_2$	300	2.8	1.8 (1.3)	2.82	1.3	3.9
	2100		2.0 (1.3)	2.79	1.8	4.8
$\text{Cl} + \text{NO} \rightarrow \text{ClNO}$	300	5.8	6.2 (0.8)	7.01	6.1	7.5
	2100		5.2 (0.8)	6.40	5.7	7.6
$\text{O} + \text{O}_2 \rightarrow \text{O}_3$	300	0.28	0.25 (0.35)	0.32	0.12	1.3
	2100		0.57 (0.35)	0.67	0.40	2.0

The agreement for the normal cases is very satisfactory, being in the range $\pm 50\%$. In view of the crude simplifications made in the present model, no better results can be expected. (The larger discrepancies for the O_3 system are due to the anomalously small α/β value of 0.076, whereas α/β in all other cases are in the range 0.3 to 0.6; see also Ref. 25. In order to facilitate the application of the present model, the calculation for the $C_2H_6 \rightleftharpoons 2CH_3$ system is demonstrated in Appendix B in detail.

In conclusion, we have presented a simplified statistical adiabatic channel model of unimolecular reactions, in which the energy pattern of the channel maxima has been analyzed and represented by empirical relations. Coupling of angular momenta, which was neglected in the initial separation of coordinates, has been corrected for *a posteriori*. The model conveniently interpolates between rigid and loose activated complex theory and uses the same language. The two interpolation parameters are the Morse parameter β and the looseness parameter α which both reflect different properties of the potential energy surface. In the present work, the limiting high pressure rate constant k_∞ was expressed in such a simple way that experimentalists can predict it with very small calculational effort (on a pocket calculator). The energy pattern of the channel maxima described in this work can immediately also be used to obtain specific rate constants $k(E, J)$ in a way which is consistent with k_∞ . This will be shown in a subsequent article.²⁴

It should be emphasized again that the present model uses a highly simplified characterization of the potential energy surface which may or may not be adequate enough. As soon as *ab initio* calculations of the surface for large interfragment distances are available, and are analyzed with respect to the channel eigenvalues, the present approach would again allow one to incorporate the energy pattern of the channel threshold energies in unimolecular rate theory in a most economic way.

ACKNOWLEDGMENTS

It is pleasure to thank M. Quack for many stimulating discussions of this work. Financial support via the Sonderforschungsbereich Photochemie mit Lasern of the Deutsche Forschungsgemeinschaft is also gratefully acknowledged.

rotor: $A_e(C_2H_6) \leftarrow \epsilon_K \rightarrow A_\infty(CH_3)$,
 torsional oscillator: $\nu_4(C_2H_6) \leftarrow \epsilon_4 \rightarrow A_\infty(CH_3)$,
 rocking oscillator: $\nu_9(C_2H_6) \leftarrow \epsilon_9 \rightarrow B_\infty(CH_3)$, degeneracy 2,
 rocking oscillator: $\nu_{12}(C_2H_6) \leftarrow \epsilon_{12} \rightarrow B_\infty(CH_3)$, degeneracy 2.

It is very important to correlate the respective reactant and product quanta in the order of increasing energies. The values of the quanta are $\nu_4(C_2H_6) = 289 \text{ cm}^{-1}$, $\nu_9(C_2H_6) = 821 \text{ cm}^{-1}$, $\nu_{12}(C_2H_6) = 1206 \text{ cm}^{-1}$, $A_e(C_2H_6) = 2.671 \text{ cm}^{-1}$, $A_\infty(CH_3) = 4.789 \text{ cm}^{-1}$, and $B_\infty(CH_3)$

APPENDIX A: ANGULAR MOMENTUM COUPLING CORRECTIONS $F_{AM_e}^*$

(a) For linear reactant molecules, there is no angular momentum coupling neglected, such that $F_{AM_e}^* = 1$.

(b) For nonlinear reactant molecules, the angular momentum coupling $|\tau_n| \leq J$ (or $|K| \leq J$ for a symmetric top) has to be accounted for. In the rigid limit $\alpha/\beta = 0$, the summations over J and K (for a prolate symmetric top) in Eq. (5.1) cannot be separated, i.e., instead of a product $Q_{\text{cent}}^* Q_K^*$, one has correctly

$$Q_{\text{cent}, K}^* = \sum_{J=0}^{\infty} (2J+1) \exp \left[-\frac{E_0(J) - E_0}{kT} \right] \times \sum_{K=-J}^+ \exp \left[-\frac{(A_e - B_e)K^2}{kT} \right].$$

If $E_0(J) - E_0$ is represented by Eq. (5.3) with $\nu \approx 1$, this expression, after replacing the sums by integrals, can be evaluated to give

$$Q_{\text{cent}, K}^* \approx \sqrt{\frac{\pi(kT)^3}{(A_e - B_e + C_\nu)C_\nu}} = \sqrt{\frac{\pi kT}{A_e - B_e + C_\nu}} Q_{\text{cent}}^*.$$

The calculation of $Q_{\text{cent}, K}^*$ by Eqs. (6.1) and (6.6) would have given

$$Q_{\text{cent}, K}^* = \Gamma\left(\frac{3}{2}\right) \sqrt{\frac{kT}{A_e}} Q_{\text{cent}}^*.$$

Therefore, the angular momentum coupling correction $F_{AM_e}^*$ in this case is equal to

$$F_{AM_e}^* = 2 \sqrt{\frac{A_e}{A_e - B_e + C_\nu}}.$$

For $\nu \neq 1$, the calculation of $F_{AM_e}^*$ is somewhat more elaborate. For a prolate symmetric top often $C_\nu < B_e \ll A_e$, such that $F_{AM_e}^* \approx 2$ is a good approximation. Since F_{AM}^* , for normal α/β values, is close to $F_{AM_\infty}^*$, the exact value of $F_{AM_e}^*$ is only of minor importance.

APPENDIX B: APPLICATION OF THE PRESENT MODEL TO THE $C_2H_6 \rightleftharpoons 2CH_3$ REACTION

For C_2H_6 , the number b of disappearing "oscillators" (including the K rotor), which correlate with the rotors of the two CH_3 fragments, is equal to six. We use for these six coordinates the correlation

$= 9,578 \text{ cm}^{-1}$. The average quantum of the five oscillators ϵ_e is equal to $\epsilon_e = 869 \text{ cm}^{-1}$; the average quantum of the six product rotors is $B_\infty = 7.982 \text{ cm}^{-1}$. [These values are used in Eqs. (3.20), (3.12), (3.23), and (3.24) only.] For the looseness parameter α we use⁴

the "standard" value $\alpha = 1.0 \text{ \AA}^{-1}$, for the Morse parameter the value $\beta = 1.8 \text{ \AA}^{-1}$, and for the dissociation energy $D = 33\,859 \text{ cm}^{-1}$.

The effective Morse parameter β_{eff} [Eq. (3.26)] is derived with $b' = 4$, $s = 18$, $\langle E_{\text{cont}}(q_e) \rangle = kT$, and $E' = D + \langle U_{\text{vib}} \rangle$. From tabulated thermodynamic data,²⁶ $\langle U_{\text{vib}} \rangle = 167 \text{ cm}^{-1}$ at 300 K and 6055 cm^{-1} at 1300 K; similar values would have roughly been guessed directly from the vibrational frequencies⁴ of C_2H_6 . With these data, $\beta_{\text{eff}}(300 \text{ K}) = 1.139 \beta$ and $\beta_{\text{eff}}(1300 \text{ K}) = 1.186 \beta$ follows from Eq. (3.26), such that the exponent x in Eq. (3.18) is given by $x = 0.501$ at 300 K and $x = 0.505$ at 1300 K [with $C_2 = 1.75$ and $n = 2.30$ from Eqs. (3.20) and (3.21), respectively], whereas $(\epsilon - B_\infty)/(\epsilon_e - B_\infty)$ in Eq. (3.22) is given by 5.23×10^{-3} at 300 K and 7.89×10^{-3} at 1300 K (with $C_3 = 8.39$ and $C_4 = 20.5$ at 300 K and $C_3 = 8.28$ and $C_4 = 20.0$ at 1300 K). For the six disappearing oscillators, the effective quanta ϵ in Eq. (3.22) hence are given by $\epsilon_K = 4.778 (4.772) \text{ cm}^{-1}$, $\epsilon_4 = 6.275 (7.031) \text{ cm}^{-1}$, $\epsilon_9 = 13.822 (15.980) \text{ cm}^{-1}$, and $\epsilon_{12} = 12.835 (19.018) \text{ cm}^{-1}$ for 300 K (1300 K). With $\Gamma(1.500)$, $\Gamma(1.501)$, and $\Gamma(1.505) \approx 0.886$, the evaluation of the partition functions Q_m^* from Eq. (6.6), then, is straightforward.

The symmetry number σ^* , with $\sigma_1 = \sigma_2 = \sigma(\text{CH}_3) = 6$, $\sigma = \sigma(\text{C}_2\text{H}_6) = 18$ (torsion levels threefold degenerate), $S_\infty = 2$, and $S_e = 1$, follows from Eq. (4.5) as 71. The angular momentum correction F_{AM}^* by Eq. (6.7) with $F_{\text{AM},e}^* \approx 2$ and $F_{\text{AM},\infty}^* = 2^6/\pi^2$ is given by Eq. (6.4). The zero point barrier is evaluated from the maximum of the potential (3.2) with $\Delta E_e = 2635 \text{ cm}^{-1}$ and $\alpha/\beta = 0.556$ (where $\epsilon_{\text{RC}} = \nu_3 = 995 \text{ cm}^{-1}$). One obtains $\Delta E_{0e} = 9.7 \text{ cm}^{-1}$, whereas the approximate relation (3.4) would have given $\Delta E_{0e} \approx 8.5 \text{ cm}^{-1}$.

Centrifugal barriers are determined with a quasi-diatomic model such that $a_1 = 2/\beta q_e = 0.73$ and $a_2 = (\beta q_e)^{-2} = 0.13$ in Eq. (5.2). With $\alpha/\beta = 0.556$, $\Delta E_e = 2635 \text{ cm}^{-1}$, $D = 33\,859 \text{ cm}^{-1}$, and $B_e = 0.6631 \text{ cm}^{-1}$, the maximum of $V_a(z)$ in Eq. (5.2) for varying z now is derived by a simple trial and error procedure for a starting value of $J = 5$ and $J = 0$. This gives $E_0(J = 5) - E_0(J = 0) \approx 1.2 \text{ cm}^{-1}$ and, hence, with $\nu \approx 1.0$, a first guess of $C_\nu^{(0)}$ of the order of 0.04 cm^{-1} . According to Eq. (5.4), then, J_{max} values near 50 for 300 K and 100 for 1300 K are estimated. The final centrifugal parameters ν and C_ν , therefore, are derived with Eqs. (5.2) and (5.3) between $J = 5$ and $J = 2J_{\text{max}} \approx 100$ for 300 K and $J = 5$ and $J = 2J_{\text{max}} \approx 200$ for 1300 K, leading to $\nu = 1.07$ and $C_\nu = 0.031 \text{ cm}^{-1}$ for 300 K and $\nu = 1.12$ and $C_\nu = 0.026 \text{ cm}^{-1}$ for 1300 K. The final calculation of the centrifugal partition function (6.3) is then straightforward. The calculation of all other factors in Eq. (7.4) is as usual: $Q_{e1}(\text{C}_2\text{H}_6) = 1$, $Q_{e1}(\text{CH}_3) = 2$, $Q_{\text{rot}}(\text{CH}_3) = \sqrt{[\pi(kT)^3/4.789] \times 9.578 \times 9.578 \text{ cm}^{-3}/6}$, and $\mu = 7.5175$

g mol^{-1} . The final results of the calculation of $k_{\text{rec},\infty}$ are included in Table VI.

- ¹J. Troe, J. Chem. Phys. **66**, 4745 (1977).
- ²J. Troe, J. Chem. Phys. **66**, 4758 (1977).
- ³J. Troe, J. Phys. Chem. **83**, 114 (1979).
- ⁴M. Quack and J. Troe, Ber. Bunsenges. Phys. Chem. **78**, 240 (1974).
- ⁵M. Quack and J. Troe, Ber. Bunsenges. Phys. Chem. **79**, 170 (1975).
- ⁶M. Quack and J. Troe, Ber. Bunsenges. Phys. Chem. **79**, 469 (1975).
- ⁷E. V. Waage and B. S. Rabinovitch, Chem. Rev. **70**, 377 (1970).
- ⁸R. A. Marcus and O. K. Rice, J. Phys. Colloid Chem. **55**, 894 (1951); R. A. Marcus, J. Chem. Phys. **20**, 355, 359, 364 (1952); **41**, 2614 (1964); **43**, 2658 (1965).
- ⁹J. C. Giddings and H. Eyring, J. Chem. Phys. **22**, 538 (1954).
- ¹⁰P. J. Robinson and K. A. Holbrook, *Unimolecular Reactions* (Wiley, London, 1972); W. Forst, *Theory of Unimolecular Reactions* (Academic, New York, 1973); J. Troe, "Unimolecular Reactions. Experiments and Theories," in *Physical Chemistry. An Advanced Treatise*, Vol. VIB, edited by W. Jost (Academic, New York, 1975); S. W. Benson, *Thermochemical Kinetics*, 2nd edition (Wiley, New York, 1976).
- ¹¹D. C. Astholz, J. Troe, and W. Wieters, J. Chem. Phys. **70**, 5107 (1979).
- ¹²G. Worry and R. A. Marcus, J. Chem. Phys. **67**, 1636 (1975).
- ¹³E. E. Nikitin, Theor. Exp. Chem. **1**, 144 (1965).
- ¹⁴P. Pechukas and J. C. Light, J. Chem. Phys. **42**, 3281 (1965).
- ¹⁵W. J. Chesnavich and M. T. Bowers, in *Gas Phase Ion Chemistry*, Vol. I (Academic, New York, 1979).
- ¹⁶M. Quack and J. Troe, in *Theoretical Chemistry. Advances and Perspectives*, Vol. VIII (Academic, New York, 1981).
- ¹⁷M. Quack and J. Troe, Ber. Bunsenges. Phys. Chem. **81**, 329 (1977).
- ¹⁸G. P. Smith and D. M. Golden, Int. J. Chem. Kinet. **10**, 489 (1978).
- ¹⁹H. S. Johnston, *Gas Phase Reaction Rate Theory* (Ronald, New York, 1966).
- ²⁰H. Gaedtke and J. Troe, Ber. Bunsenges. Phys. Chem. **77**, 24 (1973).
- ²¹D. L. Bunker and M. Pattengill, J. Chem. Phys. **48**, 772 (1968); W. H. Wong and R. A. Marcus, J. Chem. Phys. **55**, 5625 (1971).
- ²²W. L. Hase, J. Chem. Phys. **64**, 2442 (1976) and calculations cited in Ref. 23.
- ²³B. C. Garrett and D. G. Truhlar, J. Chem. Phys. **70**, 1593 (1979).
- ²⁴J. Troe, J. Chem. Phys. (to be published).
- ²⁵We have found larger discrepancies between the present model and maximum free energy calculations for large molecules (like $\text{C}_7\text{H}_8 \rightleftharpoons \text{C}_7\text{H}_7 + \text{H}$), where B_∞/ϵ_e becomes very small. For such cases, we have so far made no full statistical adiabatic channel calculation to decide which approach is more appropriate.
- ²⁶A. Burcat, TAE Report No. 411, Technion, Haifa, Israel, 1980.